CAI 5724 AI Design Project Guidelines

Overview

The AI Design Project is a hands-on creative collaboration in which **teams of 2-4 students** will develop an interactive prototype of an AI system specifically designed to address a real-world problem or opportunity in biomedical and health sciences. Teams are free to select any health-related problem that interests them, as long as the opportunity for AI to address its unmet needs are clearly rationalized. Successful completion of this project will involve the application of hands-on skills, tools, platforms, and other techniques explored throughout the course.

This project is intended to be a hands-on introduction to the fundamental tools, processes, infrastructure, and methods associated with developing custom AI systems for health. Teams are not expected to produce novel or groundbreaking results in less than 3 months. However, each team's project should be unique in some way from existing work.

The main components of the project are listed below:

- **AI Prototype** (the AI tool/system)
    - A proof-of-concept of the AI system your team developed to address the problem.
    - The form of each team's AI prototype will depend on what they proposed (e.g., an AI model, a platform, a library, an application, a chatbot, a set of agentic AI tools, etc.
    - The design of your AI prototype (i.e., what it does, and how it does it) should be clearly motivated by the specific problem your team has identified.
- **Interactive Demo** (the interactive content students will access during the presentation)
    - An interactive, shareable demonstration of your AI Prototype which your team will use during the in-class presentation.
    - The demo should showcase how your AI system works and how it can address the problem.
    - The format of the demo should either be:
        - An interactive tutorial or walkthrough document or application (e.g., Jupyter, marimo, Streamlit, or a custom web application) which is hosted online and accessible by public URL.
        - A structured, interactive in-class activity in which students use the AI Prototype (e.g., hypothetical/fictional scenarios, group exercises/activities, discussion questions, etc.). The AI Prototype must be accessible via public URL.
    - Demos should be designed to complete in approximately 10 minutes.
    - Interactive Demos will be maintained and submitted using GitHub.

- **In-Class Presentation** (on your team's selected presentation date)
    - Each team will have approximately 15 minutes to present their project to the class.
    - Each team can use any presentation format they like, as long as it includes the following in some form:

        - A brief discussion of the problem.
        - An overview of your AI Prototype and how it can help address the problem.
        - A hands-on class activity where students interact with your AI Prototype (see **Interactive Demo** above).
    - Your team's AI Prototype does not need to be completed by the time of your presentation. However, you should still have something that is functional.
- **Final Report** (due December 9)
    - A written report (approximately 2-4 pages) which documents the entire AI design and development process.
    - Each team may structure their report how they prefer, as long as it includes the following:
        - An introduction to the problem (including its background, significance, and implications if it were solved) and how AI can potentially help address it
        - An overview of your team's AI system, including what it does, how it addresses the problem, how it would be used in practice, who would use it, and how it would change the current status quo.
        - A description of the datasets your team used in the project, including the methodology for creating any synthetic or simulated data.
        - A description of any AI models or AI tools your team used in the project.
        - A narrative description of your team's overall development process, including how you built each component of the system and how you responded to any challenges or setbacks.
        - A description of any experimental results and findings.
        - A description of how your team used AI to assist in the design and development of the project.
        - A brief, high-level summary of each team member's contributions.

Grading Rubric

The AI Design Project accounts for 60% of the course grade. It will be scored out of a possible maximum of 100 points according to the rubric below:

Category	Scoring Criteria	Points
Weekly Milestones (60%)	Completion and on-time submission	40
	Milestone completion shows clear progress	10
	Response to instructor feedback	10
Final AI Prototype (10%)	Works as expected and without errors	2
	Functionality is clearly designed to address specific problem	2
	Appropriate complexity	2
	All user-created code and files for prototype on GitHub	2
	Clear improvement between midterm and final prototypes	2
Final Interactive Demo (10%)	Runs without error and is accessible by all students	2
	Background and significance of the problem are clear	2
	Uses team's real AI prototype in appropriate context with examples	2
	All user-created code and files for demo submitted	2

	Encourages user exploration (e.g., trying different inputs or settings)	2
In-Class Presentation (10%)	Clear narrative that is understandable to mixed audiences	2
	Rationale for AI system design is clearly explained	2
	Correct understanding of how the system works	2
	Class engagement and response to questions	4
Final Report (10%)	Complete, organized, readable, and accessible	2
	Appropriately revised from milestone draft versions	2
	Use of appropriate evaluation metrics and correct interpretation of results and implications	2
	Realistic assessment of limitations and drawbacks	2
	Identification of opportunities and logical next steps	2
Total Score		**100**

Project Milestones

The AI Design Project will be developed incrementally through weekly milestone assignments submitted on Canvas. Completion and on-time submission of weekly milestones accounts for 60% of the overall project grade.

The tentative schedule of milestones is listed below:

Date	Project Milestone	% of Project Grade
Sep 9	Problem Statement and AI Sketch	15%
Sep 30	AI Solution Architecture	15%
Oct 9	Development Roadmap	15%
Oct 28	Project Check-In	15%
Nov 4 – Dec 2	In-Class Presentation	20%
Dec 9	Final Report	20%

GitHub

Each team is recommended to create and maintain a GitHub repository which contains all team-created project files, including code, prompts, instructions, or other relevant files for both the AI Prototype and Interactive Demo. Do not upload large datasets and protected or sensitive files to GitHub; instead, provide directions on how to access those files. Each team's AI Prototype and Interactive Demo should be able to be recreated from files and/or directions contained in their GitHub repository.

Changes in Direction

Pivots are a normal part of AI development, and refinements and improvements based on preliminary results are welcome and expected. Once a particular milestone assignment has been submitted, there is no need to re-submit a version based on any new project directions. However, all components of the final report should be based on the final version of the project.

AI Assistance

The use of agentic AI as a collaborative team member is encouraged, especially to help write code, debug technical issues, provide real-time project feedback, or to explain technical or domain concepts. However, these tools are not without limitations, and ultimately you are responsible for all submitted content. Each team will document and reflect on the contribution of AI agents in their projects, including the strategies they found to be most effective. Teams will document and reflect on their experiences using AI assistance in the AI Teaming Log.

Responsible Data Use

Projects must use publicly available, de-identified, synthetic, or instructor-approved datasets. Do not use or share any identifiable patient data or protected data from your workplace which you are not authorized to share. Document the source and provenance of all data in your final report. Ask the instructor if you are unsure whether a given dataset is appropriate.